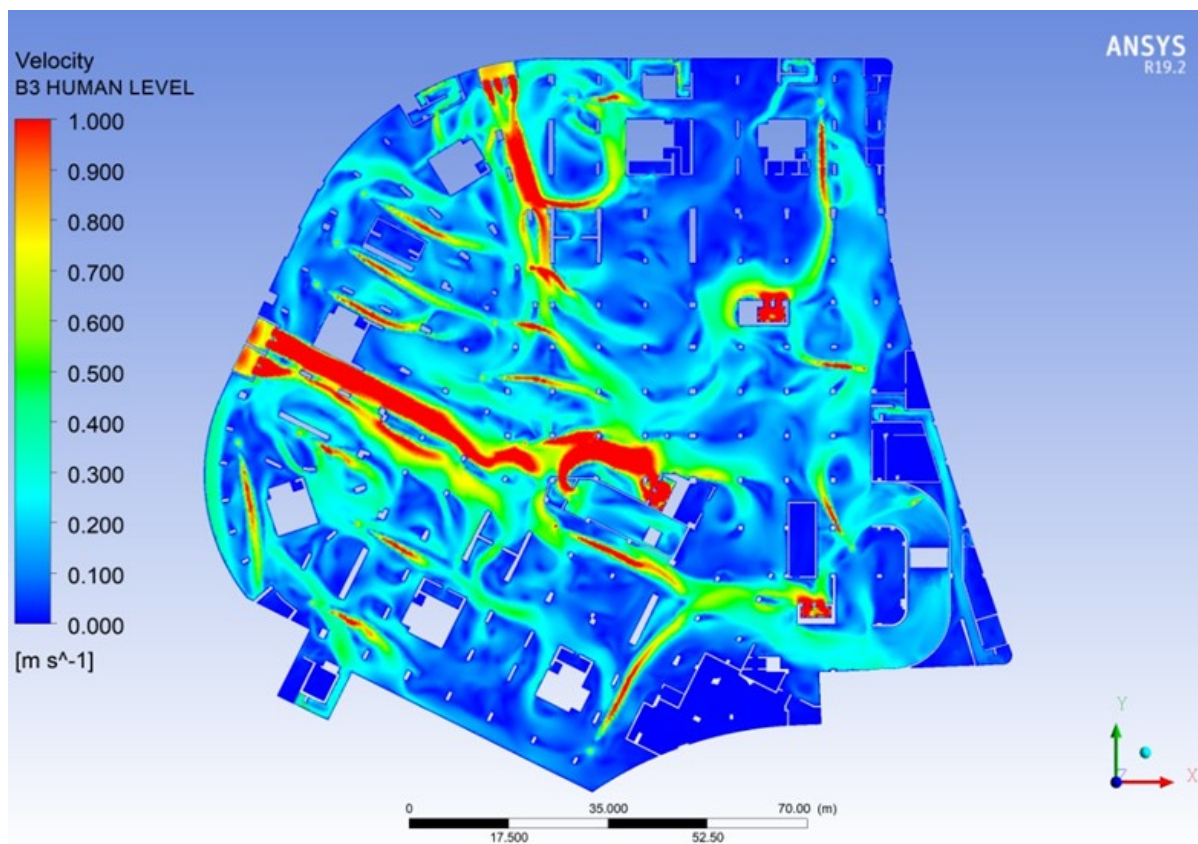


ENGINEERING ANALYSIS

CFD Assessment of Underground Car Park Ventilation and Smoke Control

Multistorey Mechanical Ventilation and Fire Scenario Study

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Human-level velocity field on Level B3

Original analysis 2021 Professional portfolio edition 2026

Executive Summary

This report documents a 2021 computational fluid dynamics assessment of the normal ventilation and fire-mode smoke-control systems in a three-level underground car park. The facility was described as serving 1,135 vehicles. The model included Levels B2, B3, and B4, 59 induction jet fans, fresh-air and extract shafts, escape routes, and fire-control devices. The analysis used ANSYS Fluent to test air movement, air-change performance, and the consequences of selected 4 MW vehicle-fire scenarios.

The central finding was a distribution problem rather than a simple lack of installed capacity. Under normal ventilation, the calculated air-change rates were 16.75 per hour on B2, 11.76 on B3, and 3.98 on B4. The project criterion was six air changes per hour. Under the fire operating condition, the corresponding results were 38.75, 27.39, and 9.12 per hour against a target of ten. B2 and B3 received considerably more airflow than required, while B4 remained below the target in both operating modes.

The velocity contours indicate that the jet fans moved air through the occupied zones and reduced stagnant regions. Their local performance did not solve the floor-to-floor imbalance because the principal supply and extract arrangement concentrated system influence near B2. The assessment therefore recommended active supply and extract fans for each floor, independently controlled smoke-extract paths, and fan locations selected to draw smoke away from escape routes while supplying replacement air toward protected egress points.

Three fire cases were evaluated with transient thermal and species-transport models. Fire controls were activated after 180 seconds in the source study. The source narrative states that two cases remained within the stated concentration limits at 90 seconds and that the B2 case approached the limit, although its floor labels conflict with the listed scenarios. By 180 seconds, the stated limits were exceeded in all cases, and the activated system did not reverse the deterioration. The study therefore did not demonstrate tenable evacuation and firefighting conditions through the modeled period.

Measure	Normal operation	Fire operation	Assessment
Target air changes	6 per hour	10 per hour	Criteria applied in the 2021 study
B2 result	16.75 per hour	38.75 per hour	Above target in both modes
B3 result	11.76 per hour	27.39 per hour	Above target in both modes
B4 result	3.98 per hour	9.12 per hour	Below target in both modes
Mesh size	22.57 million elements	Same geometry basis	Three connected floor domains
Fire source	Not applicable	4 MW per vehicle	One- and two-vehicle cases

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1 Project Definition

1.1 Purpose

The assessment tested whether the completed mechanical ventilation concept could control vehicle emissions during routine operation and manage smoke during a fire. The analysis did not redesign the building. It evaluated the supplied geometry, fans, shafts, and control assumptions independently and identified where the system failed to deliver the required floor-level performance.

Underground car parks require mechanical air movement because buoyant smoke and combustion products cannot disperse to atmosphere as they would in an open structure. The same geometry that stores vehicles efficiently can trap hot smoke, reduce visibility, and contaminate escape routes. A useful assessment therefore has to connect three questions: whether clean air reaches every part of the car park, whether smoke is directed toward effective extract points, and whether occupants retain a tenable path to safety long enough to evacuate.

1.2 Objectives

1. Verify the air-change rate and the distribution of velocity under normal ventilation.
2. Test one- and two-vehicle fire scenarios at critical locations selected from escape distances.
3. Track heat, smoke, carbon monoxide, carbon dioxide, and visibility-relevant flow conditions at occupied heights.
4. Determine whether the existing jet fans, supply paths, extract paths, and fire controls produced a tenable evacuation and firefighting environment.
5. Recommend changes to fan duties, floor separation, and equipment location where the modeled system did not meet the project criteria.

1.3 Scope and Use

The report is a reconstruction of the 2021 engineering record. The CFD cases were not rerun for this edition. Results are presented as reported by the original analysis, with inconsistent source values identified in Appendix B. The cited standards form the basis used by the study; a current design would require confirmation against the editions adopted by the authority having jurisdiction, together with a complete fire-engineering review.

2 Facility and Ventilation System

2.1 Facility Geometry

The modeled car park occupied three basement levels. Each level was divided into three zones for planning and ventilation review. The source record states a total car-park area of 42,660 m², while the listed zone areas total 31,350 m². Because the source does not explain the difference, both values are retained and the discrepancy is identified rather than reconciled by assumption.

Level	Zone 1 m ²	Zone 2 m ²	Zone 3 m ²	Listed total m ²	Clear height mm
B2	3,250	3,500	3,700	10,450	3,160
B3	3,250	3,500	3,700	10,450	2,630
B4	3,250	3,500	3,700	10,450	2,630



Ventilation layouts for Levels B4, B3, and B2

2.2 Jet Fan System

The car park used 59 Systemair IV 50-4/8 induction jet fans: 19 on B4 and 20 on each of B3 and B2. The fans were represented at two operating points. The lower-speed condition supplied 2,929 m³/h and approximately 14 N of thrust; the higher-speed fire condition supplied 5,825 m³/h and approximately 57 N. The assessment treated the jet fans as momentum devices that entrain the surrounding air and drive it toward the principal extract paths.

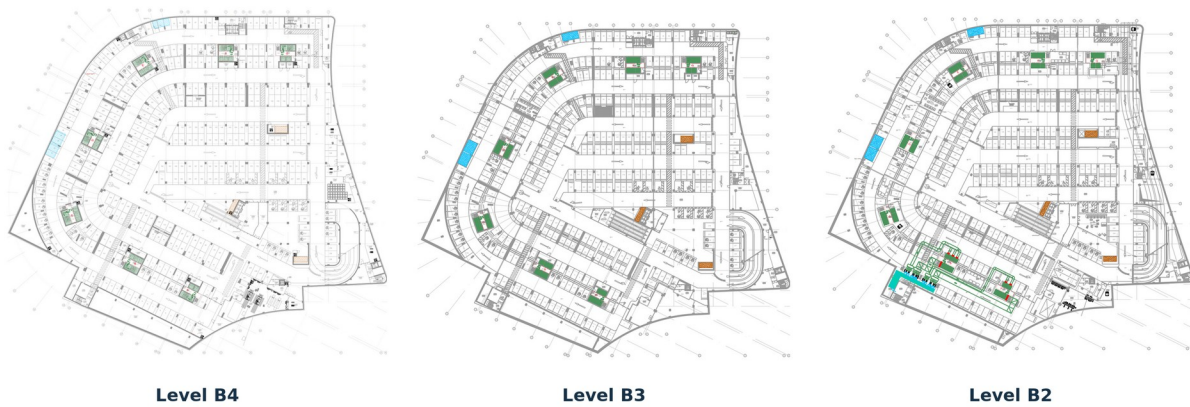
$$F = Q v \rho$$

In this expression, F is thrust, Q is volumetric flow, v is discharge velocity, and rho is air density. The jet fans redistribute air within a floor; they do not create the overall mass flow that must enter through supply paths and leave through extract shafts.

Item	Normal setting	Fire setting
Airflow per jet fan	2,929 m ³ /h	5,825 m ³ /h
Nominal shaft power	0.37 kW	1.64 kW
Impeller speed	738 r/min	1,460 r/min
Thrust	14 N	57 N

2.3 Supply Extract and Pressurization

The shaft schedule included three fresh-air paths and three extract paths. The modeled fire duties approximately doubled the normal shaft flows. The original configuration placed the principal fire-fan influence on B2 and relied on openings and inter-floor paths to affect the lower levels. That arrangement became the decisive weakness of the system because the available airflow was not apportioned according to the demand on each floor.



Smoke-control and pressurization layouts for Levels B4, B3, and B2

3 Analysis Criteria

3.1 Design Basis

The 2021 study cited BS 7346-7:2013 for smoke and heat control in covered car parks and BS 9999:2017 for fire-safety design and management. It also referred to BR 368 and the Smoke Control Association guidance for CFD modelling. The analysis used six air changes per hour as the normal-ventilation target and ten per hour as the fire-mode target.

Performance question	Criterion used in the study	Model output
Routine air renewal	At least 6 air changes per hour	Net floor flow and calculated ACH
Fire-mode air renewal	At least 10 air changes per hour	Net floor flow and calculated ACH
Air distribution	No materially stagnant occupied zones	Velocity contours at three elevations
Escape-path exposure	CO below 50 ppm and CO ₂ below 5,000 ppm in the stated review	Transient species concentrations
Smoke movement	Direct smoke toward effective extract paths and preserve egress	Transient velocity, temperature, and species fields

3.2 Air Change Calculation

Air changes per hour were calculated from the net volumetric flow through each floor and the corresponding modeled volume.

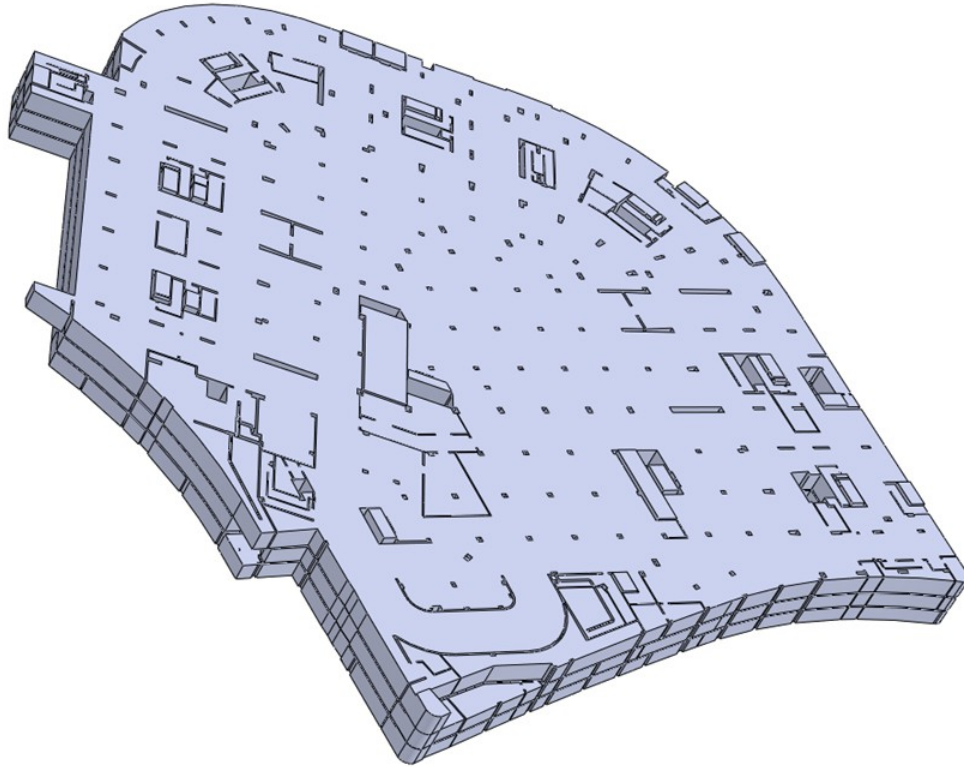
$$ACH = \frac{3600 Q}{V}$$

Here, Q is the net flow in cubic metres per second and V is the modeled floor volume in cubic metres. The factor 3,600 converts seconds to hours. This calculation checks total renewal; velocity contours are still required because an acceptable average can conceal local dead zones or short-circuiting between supply and extract points.

4 CFD Model and Numerical Method

4.1 Geometry Preparation

The architectural and mechanical plans were converted into an inverse fluid domain representing the air volume. Walls, columns, ramps, cores, rooms, shafts, vents, jet fans, and selected fire locations were retained where they affected the flow. Small details that did not materially influence the solution were simplified so computational effort could be concentrated around openings, equipment, and occupied paths.

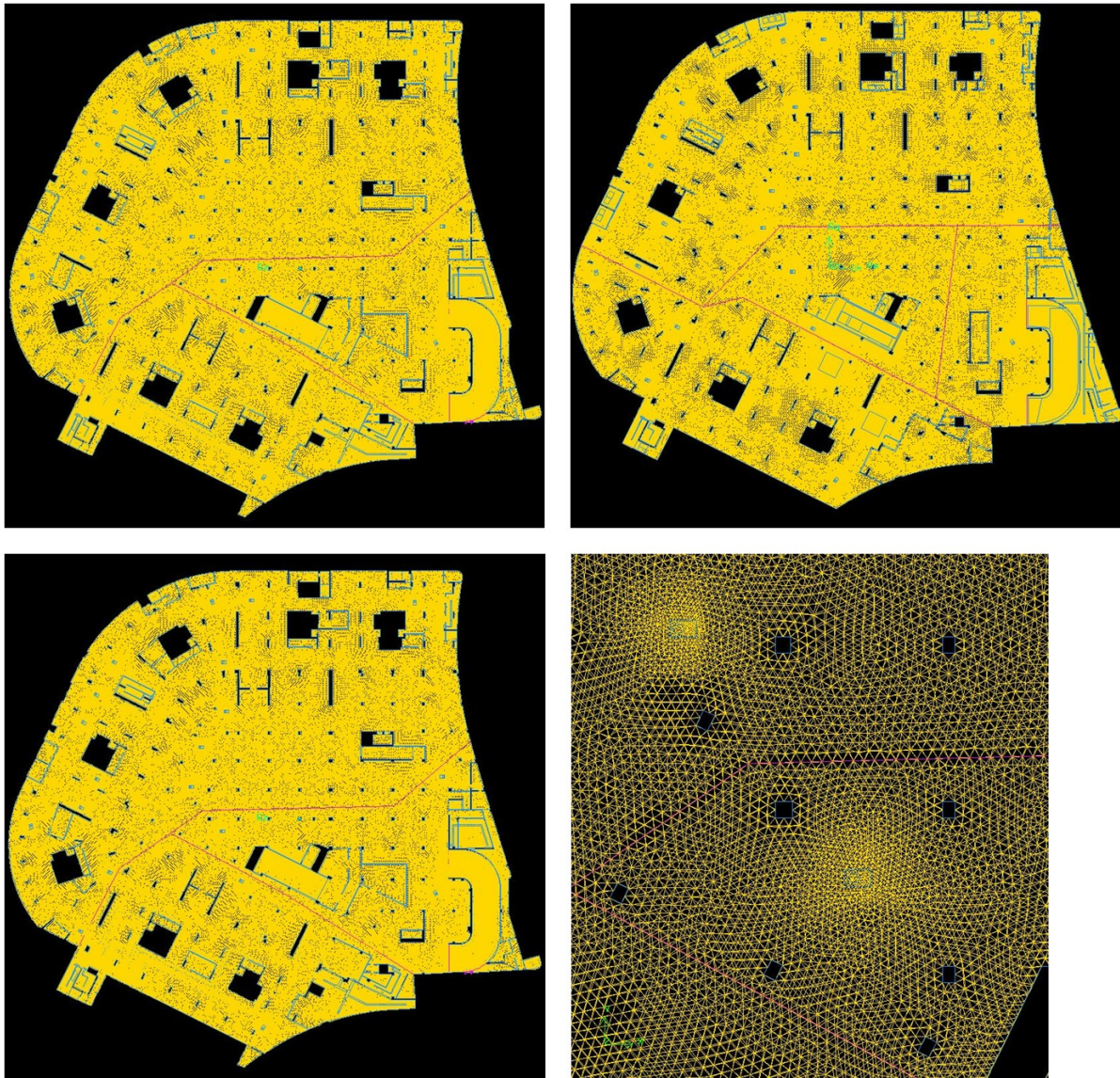


Inverse three-dimensional fluid domain used for the CFD model

4.2 Mesh Strategy

The three domains contained approximately 4.19 million nodes and 22.57 million elements. The nominal element size ranged from 50 mm near inlets and other sensitive features to 500 mm in regions where a coarser mesh was considered adequate. Local refinement was applied around vents, jet-fan outlets, fire locations, and geometric changes where higher velocity or pressure gradients were expected.

Domain	Nodes	Elements	Share of elements
B2	1,512,777	8,192,349	36.3%
B3	1,481,166	8,002,797	35.5%
B4	1,194,084	6,373,650	28.2%
All domains	4,188,027	22,568,796	100.0%



Floor meshes and local refinement around flow-critical features

4.3 Solver Configuration

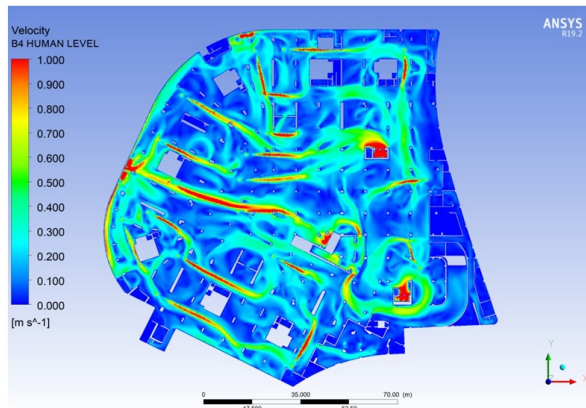
Normal ventilation was treated as a steady-state flow problem using the standard k-epsilon turbulence model with standard wall functions. The fire cases were transient. Gravity and ideal-gas density were enabled to represent buoyancy, the energy equation was solved for heat transfer, and species transport was used to follow combustion products. The source record states that the simulations ran on a 32-core HP server in ANSYS Fluent.

Analysis	Time treatment	Physics	Primary outputs
Normal ventilation	Steady state	Incompressible airflow, standard k-epsilon turbulence	Velocity and floor flow balance
Fire and smoke control	Transient	Gravity, ideal-gas density, energy, turbulence, species transport	Smoke path, temperature, CO, CO ₂ , and time

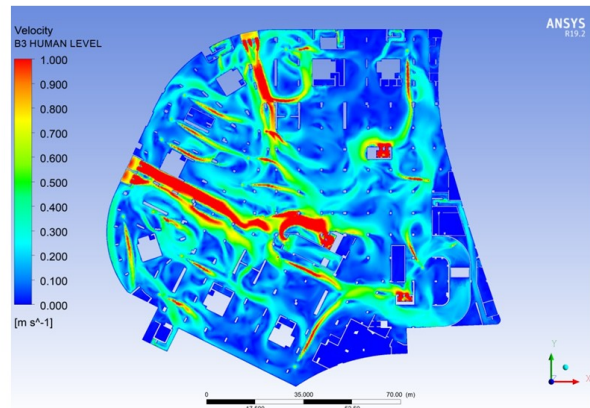
5 Normal Ventilation Results

5.1 Velocity Review

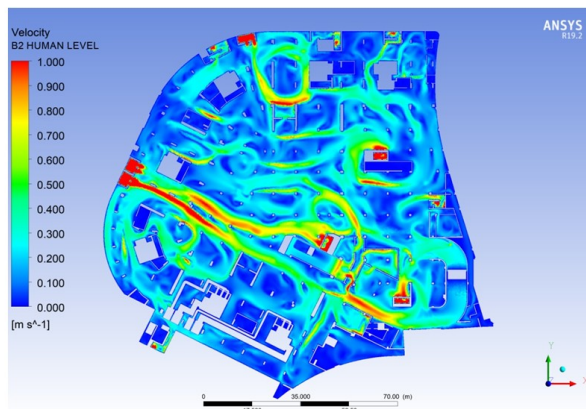
Velocity was reviewed at waist level, human level, and jet-fan level. Each plane was plotted with maximum legend values of 6, 3, and 1 m/s. The repeated scales were necessary because a 6 m/s legend shows the jet discharge but can hide low-speed circulation, while the 1 m/s legend makes weak paths and stagnant pockets visible.



B4 human level



B3 human level



B2 human level

Human-level velocity contours on B4, B3, and B2 using the same 0 to 1 m/s scale

The contours show that the jet fans generated recognizable circulation paths across all three floors. This supports the source conclusion that the local jet-fan duties were broadly effective. The plots also show that local movement alone cannot establish whether the complete floor receives the required volume of replacement air. That question is answered by the mass-flow balance and ACH calculation.

5.2 Floor Flow Balance

Level	Net flow m ³ /s	Volume used m ³	Calculated ACH	Target ACH	Result
B2	181.08	38,922.61	16.75	6.00	Pass
B3	107.52	32,909.21	11.76	6.00	Pass
B4	36.27	32,819.01	3.98	6.00	Fail

B4 achieved only 66 percent of the normal-ventilation target. At the same time, B3 delivered almost twice the target and B2 delivered approximately 2.8 times the target. The result points to hydraulic imbalance across the connected floors. Increasing every fan equally would add energy and velocity without addressing why the lowest level receives the smallest share of the available airflow.

5.3 Normal Ventilation Decision

The jet fans should be retained as circulation devices, subject to final placement review. The corrective action should focus on the sources and sinks of the floor flow: provide active supply and extract duty at each level, balance the pressure paths, and commission each floor against measured airflow. The design should be adjusted until B4 reaches the criterion without maintaining unnecessary excess on B2 and B3.

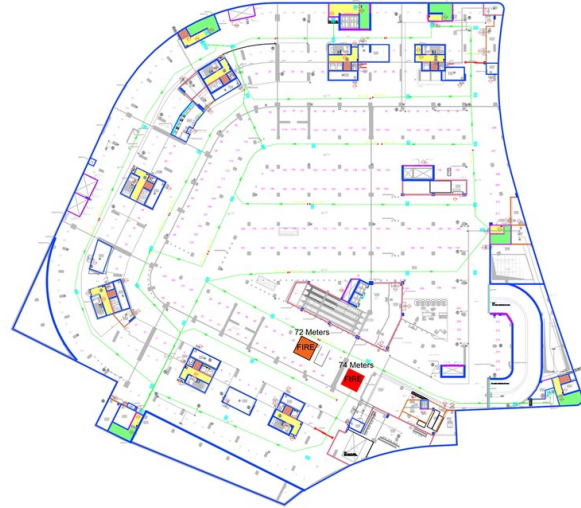
6 Fire and Smoke Control Analysis

6.1 Scenario Selection

The fire locations were selected from long escape paths and areas expected to challenge smoke extraction. The two B4 locations were approximately 72 m and 74 m from the identified safe points. The model included one B4 vehicle fire, two simultaneous B4 vehicle fires, and one B2 vehicle fire. Fire-mode jet fans, supply paths, extract paths, and curtains were activated after 180 seconds in the original control sequence.



Level B4 fire locations



Level B2 fire location

Selected fire locations and escape-path distances on B4 and B2

Case	Active fire source	Purpose
1	One 4 MW vehicle fire on B4	Test a critical lower-level escape path
2	Two simultaneous 4 MW vehicle fires on B4	Test the most severe lower-level source case
3	One 4 MW vehicle fire on B2	Test the level closest to the principal fire-fan influence

6.2 Fire Source

Each vehicle fire was represented by a 4 MW heat-release rate over a 2 m by 5 m area with a 14 m perimeter, consistent with the source study's interpretation of the sprinklered-car-park design case. Polyurethane was used as the fuel surrogate with a heat of combustion of 26,200 kJ/kg. The resulting fuel mass-flow rate was approximately 0.153 kg/s per fire.

$$\dot{m} = \frac{\dot{Q}}{\Delta H_c}$$

Parameter	Value used
Heat-release rate	4 MW per vehicle
Fire footprint	2 m by 5 m
Fuel surrogate	Polyurethane CH1.8O0.3N0.05
Heat of combustion	26,200 kJ/kg
Fuel mass-flow rate	0.153 kg/s
Control activation	180 seconds after ignition

6.3 Fire Mode Air Change

Level	Net flow m ³ /s	Volume used m ³	Calculated ACH	Target ACH	Result
B2	355.19	33,000	38.75	10.00	Pass
B3	209.19	27,500	27.39	10.00	Pass
B4	69.67	27,500	9.12	10.00	Fail

The fire-mode calculation repeated the normal-mode pattern. B2 and B3 exceeded the target by large margins, while B4 reached approximately 91 percent. The values confirm that the installed system could move a substantial total volume of air. They also confirm that the fire operating sequence did not distribute that capacity according to the risk on each floor.

6.4 Tenability Review

The source study compared evacuation travel with conditions along the selected paths. It reported that an able occupant could reach an exit in about 60 seconds under the assumed movement speed, while delayed detection or reduced mobility could extend evacuation to 180 seconds. Its narrative states that two cases remained within the stated CO and CO₂ limits at 90 seconds and that the B2 case was at or near the boundary, but the floor labels do not match the scenario list. By 180 seconds, the reported limits were exceeded in all cases.

The controls activated at 180 seconds did not reduce the fire effects quickly enough to recover the modeled environment. The study therefore did not establish a tenable period for delayed occupants or a reliable approach path for firefighters. This conclusion should be read with the limitations of the available record: the complete time-history plots, visibility thresholds, sprinkler response, leakage assumptions, and sensitivity cases were not preserved in the public summary.

7 Engineering Diagnosis

7.1 The Governing Failure

The governing failure was not jet-fan thrust. The velocity plots show that the fans created local momentum and circulation. The governing failure was how the building-level supply and extract duties were connected to the three floors. B2 sat closest to the principal mechanical influence and received the highest flow. B4 depended on the same shared system but received the lowest flow, even though it contained the longest modeled escape paths and the most demanding two-fire case.

Observation	Evidence	Engineering meaning
Local circulation exists	Velocity paths visible around jet-fan routes	Jet fans perform their intended momentum function
B2 is heavily over-supplied	16.75 ACH normal and 38.75 ACH fire	Available capacity is concentrated near the upper basement
B4 remains deficient	3.98 ACH normal and 9.12 ACH fire	Shared shafts do not deliver sufficient floor-level flow
Late activation does not recover conditions	All reported cases exceed stated limits by 180 s	Control timing and smoke capture require redesign
One-floor fire-fan influence is ineffective	B3 and B4 depend on indirect paths	Each floor needs an intentional supply and extract strategy

7.2 Decision Path

1. Check the local velocity field. The jet fans create circulation, so replacing them is not the first corrective action.
2. Close the mass balance by floor. B2 and B3 exceed the targets while B4 fails, which identifies a distribution problem.
3. Test the fire sequence at critical escape locations. The selected B4 paths control the evacuation risk.
4. Compare available safe time with evacuation demand. The 180-second adverse case is not protected by the modeled control sequence.
5. Move the design decision upstream. Provide floor-specific supply, extract, controls, and commissioning targets before refining individual jet-fan positions.

8 Recommendations and Development Path

8.1 System Recommendations

1. Provide independently driven fresh-air and extract duty on each basement level, sized to meet the required floor flow in both normal and fire operation.
2. Locate smoke extract points to capture the predicted smoke path and locate replacement-air points so they support, rather than contaminate, escape routes.
3. Rebalance B2 and B3 instead of preserving their large excess airflow while adding capacity to B4.
4. Review detection and control timing. The modeled 180-second activation occurs after the adverse evacuation case has already developed.
5. Retain jet fans where contour review confirms useful circulation, then optimize their direction and spacing after the principal floor flows have been corrected.
6. Commission the completed system with measured shaft flow, pressure, and velocity checks on every level and compare those measurements with the CFD boundary conditions.

8.2 Analysis Development

A design-stage CFD update should include a mesh-sensitivity study, explicit convergence criteria, leakage and door-position assumptions, sprinkler activation and heat-release-rate sensitivity, fan failure cases, and more than one wind or external-pressure condition. The analysis should report visibility, temperature, radiant exposure, CO, and fractional effective dose along time-dependent evacuation and firefighter access paths.

The model should also separate available safe egress time from required safe egress time. Detection, recognition, pre-movement, walking speed, queuing, mobility impairment, and stair or lobby pressurization should be treated as explicit inputs rather than combined into a single assumed evacuation duration.

8.3 Conclusion

The 2021 assessment established a clear engineering diagnosis. The car park had substantial fan and shaft capacity, and the jet fans generated circulation, but the connected system did not deliver that capacity where it was needed. B4 failed the normal and fire ACH checks and the modeled fire sequence did not preserve the stated tenability limits through 180 seconds. The appropriate response is a floor-specific supply and extract strategy, revised control timing, and validation against a complete fire-engineering performance specification.

References

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6. Savas, E. Smoke Extraction Modelling Using Computational Fluid Dynamics Software Fluent in Case of Fire Emergency at a Workplace. Ankara, 2015.
7. Balik, G., Beceren, K., and Kilic, A. Tesisat Muhendisligi Dergisi, issue 80, 2004.
8. NBN S 21-208-2. Fire protection in buildings. Design and calculation of smoke and heat extraction installations. Covered car parking buildings. 2006.

Appendix A Key Input Data

A.1 Main Shaft Boundary Conditions

Boundary	Normal flow m ³ /h	Fire flow m ³ /h
Fresh air GSF-1	90,000	180,000
Fresh air GSF-2	90,000	180,000
Fresh air GSF-3	80,000	160,000
Extract EX-1	120,000	240,000
Extract EX-2	130,000	260,000
Extract EX-3	130,000	260,000

A.2 Combustion Source Terms

Species or term	Source value per 4 MW fire
Carbon dioxide	272.1 g/s
Water vapour	245.4 g/s
Nitrogen	6.1 g/s
Carbon monoxide	1.35 g/s
Soot carbon	20.15 g/s
Oxygen consumption	274.85 g/s
Positive product flow	545.05 g/s

A.3 Result Calculation Check

The reported air-change values reproduce the stated net flows and the volumes shown in their respective result tables. For example, B4 normal ventilation gives 36.268 m³/s multiplied by 3,600 and divided by 32,819 m³, which equals 3.98 air changes per hour. B4 fire operation gives 69.672 m³/s multiplied by 3,600 and divided by 27,500 m³, which equals 9.12 per hour.

Appendix B Source Data Reconciliation

B.1 Values Requiring Confirmation

The following items do not invalidate the principal diagnosis, but they should be resolved before the model is used for design approval or current compliance review.

Source item	Recorded values	Required resolution
Total floor area	42,660 m ² stated; listed zones total 31,350 m ²	Confirm gross area, modeled area, and areas excluded from the zone schedule
Volumes used for ACH	Normal: 38,923 / 32,909 / 32,819 m ³ ; fire: 33,000 / 27,500 / 27,500 m ³	Explain whether fire values represent smoke zones or a different active volume
Mesh figure labels	The source labels two floor meshes as B2	Confirm the intended B2, B3, and B4 mapping
Fire figure references	Scenario text and figure numbers conflict	Link each scenario to the correct floor plan and solver case
Tenability thresholds	CO and CO ₂ limits are stated; temperature and visibility criteria are not fully recorded	Restore the complete acceptance criteria and time-history plots
Fan availability	Nominal duties are listed without failure combinations	Test duty and standby arrangements, loss of power, and fan failure cases

The main conclusion remains stable across the reported volume sets: B4 is the only floor below the applied ACH target in both operating modes, while B2 and B3 exceed their targets by substantial margins. Correcting the source records would improve traceability and design confidence, but it would not change the first redesign priority identified by this assessment.